

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/24**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
24: Further Statistics 2
(Part of option G only)

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
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Advice

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Turn over ►

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1. In a singing competition, two judges ranked the vocal **quality** of each of 8 singers.

The singers are labelled *A* to *H* and the table below shows the ranks given by each judge.

Rank	1	2	3	4	5	6	7	8
Judge 1	<i>C</i>	<i>B</i>	<i>A</i>	<i>G</i>	<i>E</i>	<i>F</i>	<i>H</i>	<i>D</i>
Judge 2	<i>B</i>	<i>A</i>	<i>C</i>	<i>G</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>H</i>

- (a) Calculate the Spearman's rank correlation coefficient between the judges' ranks.

(4)

Each judge also gave a mark out of 20 for each singer's vocal **range**.

Singer	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Judge 1	18	13	16	12	11	7	13	6
Judge 2	15	17	14	14	9	8	12	8

- (b) Without carrying out any further calculations, explain how the Spearman's rank correlation coefficient between the judges' marks for vocal **range** should be calculated.

(2)



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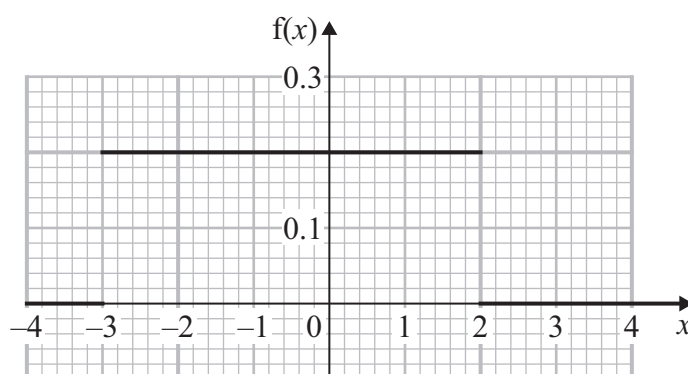
Question 1 continued

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(Total for Question 1 is 6 marks)



2. The graph of the probability density function $f(x)$ of the continuous uniform random variable X is shown below.



- (a) Write down the value of $P(X = 1)$ (1)
- (b) Find $E(X)$ (1)
- (c) Find $\text{Var}(X)$ (1)
- (d) Sketch the cumulative distribution function of X for $-3 \leq x \leq 2$
You should label any points where the sketch touches or crosses the coordinate axes. (3)
- (e) Find $P(X^2 > 1.96)$ (3)

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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 9 marks)



3. Dina is investigating the relationship between a marathon runner's body mass index (BMI), $x \text{ kg m}^{-2}$, and the runner's average speed when running a marathon, $v \text{ m s}^{-1}$

She collects data from 20 marathon runners.

Some summary statistics are given below.

$$\bar{x} = 19.8 \quad \bar{v} = 5.2 \quad \sum xv = 2056.63 \quad S_{xx} = 15.78 \quad S_{vv} = 0.94$$

- (a) Show that $S_{xv} = -2.57$ (1)
- (b) Find the value of the product moment correlation coefficient between v and x (2)
- (c) Use a suitable test, with a 5% level of significance, to assess whether or not there is evidence of a negative correlation between BMI and average speed. State the hypotheses and the critical value used. (3)
- (d) Find the equation of the regression line of v on x in the form $v = a + bx$ giving the values of a and b to 2 decimal places. (3)
- (e) Calculate the residual sum of squares (RSS). (2)

Information about 2 of the 20 marathon runners is listed in the table below.

Runner	BMI	Residual
<i>A</i>	20.9	0.18
<i>B</i>	19.7	-0.22

- (f) Determine which of these runners took the shorter amount of time to run the marathon. You must show your working. (3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

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(Total for Question 3 is 14 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 11 marks)

TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS



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Advanced Subsidiary
Further Mathematics options
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1. A continuous random variable X has cumulative distribution function $F(x)$ given by

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{5}(x+1)^2 & -1 \leq x \leq 0 \\ 1 - \frac{1}{20}(4-x)^2 & 0 < x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) Find the probability density function, $f(x)$ (2)

- (b) (i) Sketch $f(x)$ (2)

- (ii) Hence describe the skewness of the distribution. (1)

- (c) Find, to 3 significant figures, the value of c such that

$$P(1 < X < c) = P(c < X < 2) \quad (3)$$



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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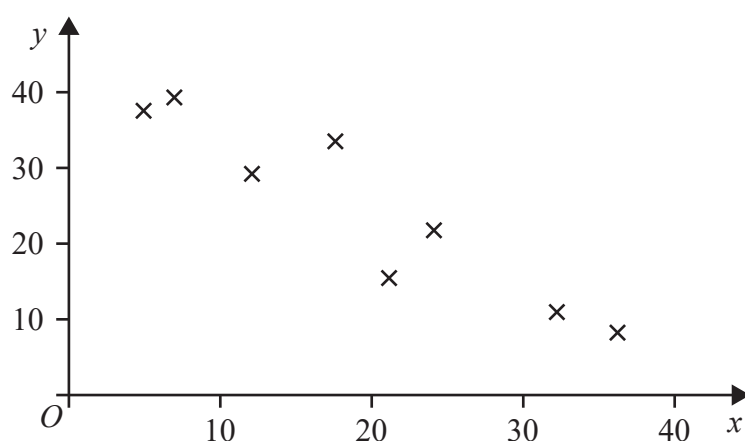
Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



2. A random sample of size $n = 8$ of paired data is taken from a population. The data are plotted below.



Test, at the 1% level of significance, whether or not there is evidence of a negative **rank** correlation between the two variables.

You should state your hypotheses and critical value and show your working clearly.

(7)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)

3. The continuous random variable Y has probability density function

$$f(y) = \begin{cases} \frac{1}{24}(y+2)(4-y) & 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the mode of Y is 1, justifying your reasoning.

(2)

Given that $P(Y < 1) = \frac{13}{36}$

- (b) determine whether the median of Y is less than, equal to, or greater than 2
Give a reason for your answer.

(2)

Given that $E(Y^2) = \frac{213}{80}$

- (c) find, using algebraic integration, $\text{Var}(2Y)$

(5)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. The continuous random variable X is uniformly distributed over the interval $[2, 7]$

(a) Write down the value of $E(X)$

(1)

(b) Find $P(1 < X < 4)$

(1)

(c) Find $P(2X^2 - 15X + 27 > 0)$

(3)

(d) Find $E\left(\frac{3}{X^2}\right)$

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 8 marks)

TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS



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Advanced Subsidiary

Further Mathematics options

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 10 marks)



2. A continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{x}{16}(9-x^2) & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the cumulative distribution function of X (3)
- (b) Calculate $P(X > 1.8)$ (2)
- (c) Use calculus to find $E\left(\frac{3}{X} + 2\right)$ (3)
- (d) Show that the mode of X is $\sqrt{3}$ (3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

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(Total for Question 2 is 11 marks)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)



4. The random variable X has a continuous uniform distribution over the interval $[-3, k]$

Given that $P(-4 < X < 2) = \frac{1}{3}$

- (a) find the value of k

(3)

A computer generates a random number, Y , where

- Y has a continuous uniform distribution over the interval $[a, b]$
- $E(Y) = 6$
- $\text{Var}(Y) = 192$

- $E(Y) = 6$

- $\text{Var}(Y) = 192$

The computer generates 5 random numbers.

- (b) Calculate the probability that at least 2 of the 5 numbers generated are greater than 7.5

(6)

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Question 4 continued

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Question 4 continued

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Paper
reference

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Further Mathematics

Advanced Subsidiary

Further Mathematics options

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(Part of option G only)

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1. Abena and Meghan are both given the same list of 10 films.

Each of them ranks the 10 films from most favourite to least favourite.

For the differences, d , between their ranks for these 10 films, $\sum d^2 = 84$

- (a) Calculate Spearman's rank correlation coefficient between Abena's ranks and Meghan's ranks.

(1)

A test is carried out at the 5% level of significance to see if there is agreement between their ranks for the films.

The hypotheses for the test are

$$H_0 : \rho_s = 0 \quad H_1 : \rho_s > 0$$

- (b) (i) Find the critical region for the test.

- (ii) State the conclusion of the test.

(2)

An 11th film is added to the list. Abena and Meghan both agree that this film is their least favourite.

A new test is carried out at the 5% level of significance using the same hypotheses.

- (c) Determine the conclusion of this test. You should state the test statistic and the critical value used.

(4)



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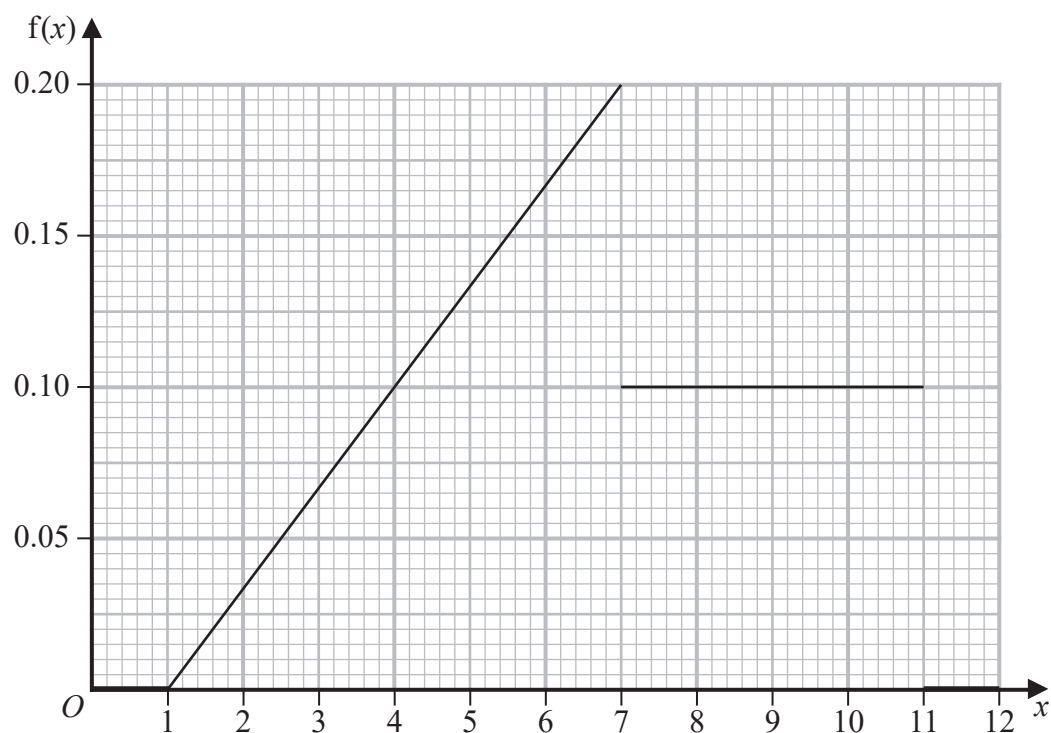
Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. The graph shows the probability density function $f(x)$ of the continuous random variable X



- (a) Find $P(X < 4)$ (2)
- (b) Specify the cumulative distribution function of X for $7 \leq x \leq 11$ (3)

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. Gabriela is investigating a particular type of fish, called bream. She wants to create a model to predict the weight, w grams, of bream based on their length, x cm.

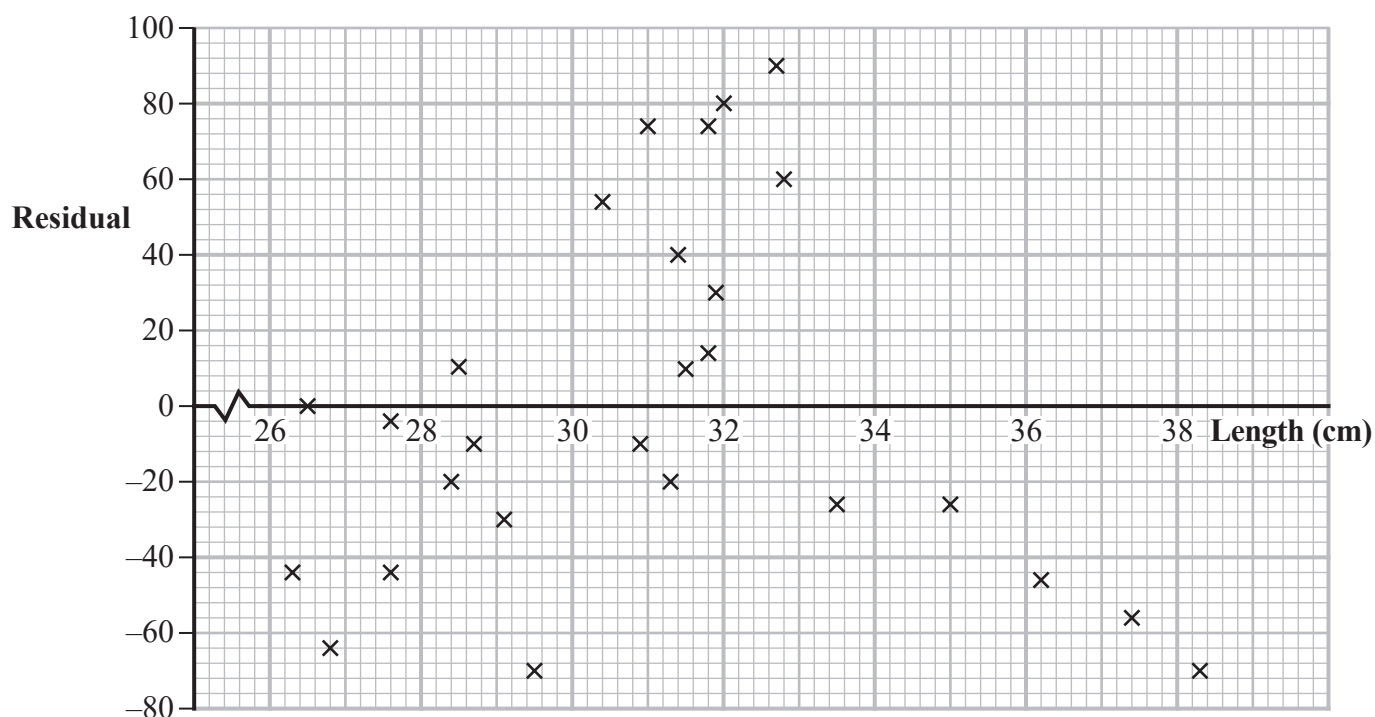
For a sample of 27 bream, some summary statistics are given below.

$$\bar{x} = 31.07 \quad \bar{w} = 628.59 \quad \sum w^2 = 11\,386\,134$$

$$S_{xw} = 13\,082.3 \quad S_{xx} = 260.8$$

- (a) Find the value of the product moment correlation coefficient between x and w (3)
- (b) Explain whether the answer to part (a) is consistent with a linear model for these data. (1)
- (c) Find the equation of the regression line of w on x in the form $w = a + bx$ (3)

A residual plot for these data is shown below.



One of the bream in the sample has a length of 32 cm.

- (d) Find its weight. (2)
- (e) With reference to the residual plot, comment on the model for bream with lengths above 33 cm. (1)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



5. The random variable X has the continuous uniform distribution over the interval $[0.5, 2.5]$

Talia selects a number, T , at random from the distribution of X

- (a) Find $P(T < 1)$
- (1)**

Malik takes Talia's number, T , and calculates his number, M , where $M = \frac{1}{T^2}$

- (b) Find the probability that both T and M are less than 2.25 (3)

Raja and Greta play a game many times.

Each time they play they use a number, R , randomly selected from the distribution of X

Raja's score is R

Greta's score is G , where $G = \frac{2}{R^2}$

- (c) Determine, giving a reason, who you would expect to have the higher **total score**. (5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS



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Level 3 GCE**

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Thursday 08 October 2020

Afternoon

Paper Reference **8FM0/24**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

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1. An estate agent in *Tornep* believes that houses further from the railway station are more expensive than those that are closer. She took a random sample of 22 three-bedroom houses in *Tornep* and calculated the product moment correlation coefficient between the house price and the distance from the station to be 0.3892

Stating your hypotheses clearly, use a 5% level of significance to test the estate agent's belief. State the critical region used in your test.

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 3 marks)



2. Mary, Jahil and Dawn are judging the cakes in a village show. They have 5 features to consider and each feature is awarded up to 5 points. The total score the judges gave each cake are given in the table below.

Cake	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>
Mary	19	17	23	10	21	15	12	8	14
Jahil	22	18	21	10	24	20	16	12	15
Dawn	9	11	6	18	9	15	13	20	13

- (a) Calculate Spearman's rank correlation coefficient between Mary's scores and Jahil's scores. (4)

- (b) Calculate Spearman's rank correlation coefficient between Jahil's scores and Dawn's scores. (3)

The judges discussed their interpretation of the points system and agreed that the first prize should go to cake *C*.

- (c) Explain how different interpretations of the points system could give rise to the results in part (a) and part (b). (2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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3. The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 4 \\ px - k\sqrt{x} & 4 \leq x \leq 9 \\ 1 & x > 9 \end{cases}$$

where p and k are constants.

- (a) Find the value of p and the value of k .

(4)

Given that $E(X) = \frac{119}{18}$

- (b) show that $\text{Var}(X) = 2.05$ to 3 significant figures.

(6)

- (c) Write down the mode of X .

(1)

- (d) Find the exact value of the constant a such that $P(X \leq a) = \frac{7}{27}$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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4. Some students are investigating the strength of wire by suspending a weight at the end of the wire. They measure the diameter of the wire, d mm, and the weight, w grams, when the wire fails. Their results are given in the following table.

	These 14 points are plotted on page 13														Not yet plotted			
d	0.5	0.6	0.7	0.8	0.9	1.1	1.3	1.6	2	2.4	2.8	3.3	3.5	3.9	4.5	4.6	4.8	5.4
w	1.2	1.7	2.3	3.0	3.8	5.6	7.7	11.6	18	25.9	34.9	47.4	52.7	63.9	81	83.6	89.9	109.4

The first 14 points are plotted on the axes on page 13.

- (a) On the axes on page 13, complete the scatter diagram for these data. (1)
- (b) Use your calculator to write down the equation of the regression line of w on d . (2)
- (c) With reference to the scatter diagram, comment on the appropriateness of using this linear regression model to make predictions for w for different values of d between 0.5 and 5.4 (1)

The product moment correlation coefficient for these data is $r = 0.987$ (to 3 significant figures).

- (d) Calculate the residual sum of squares (RSS) for this model. (2)

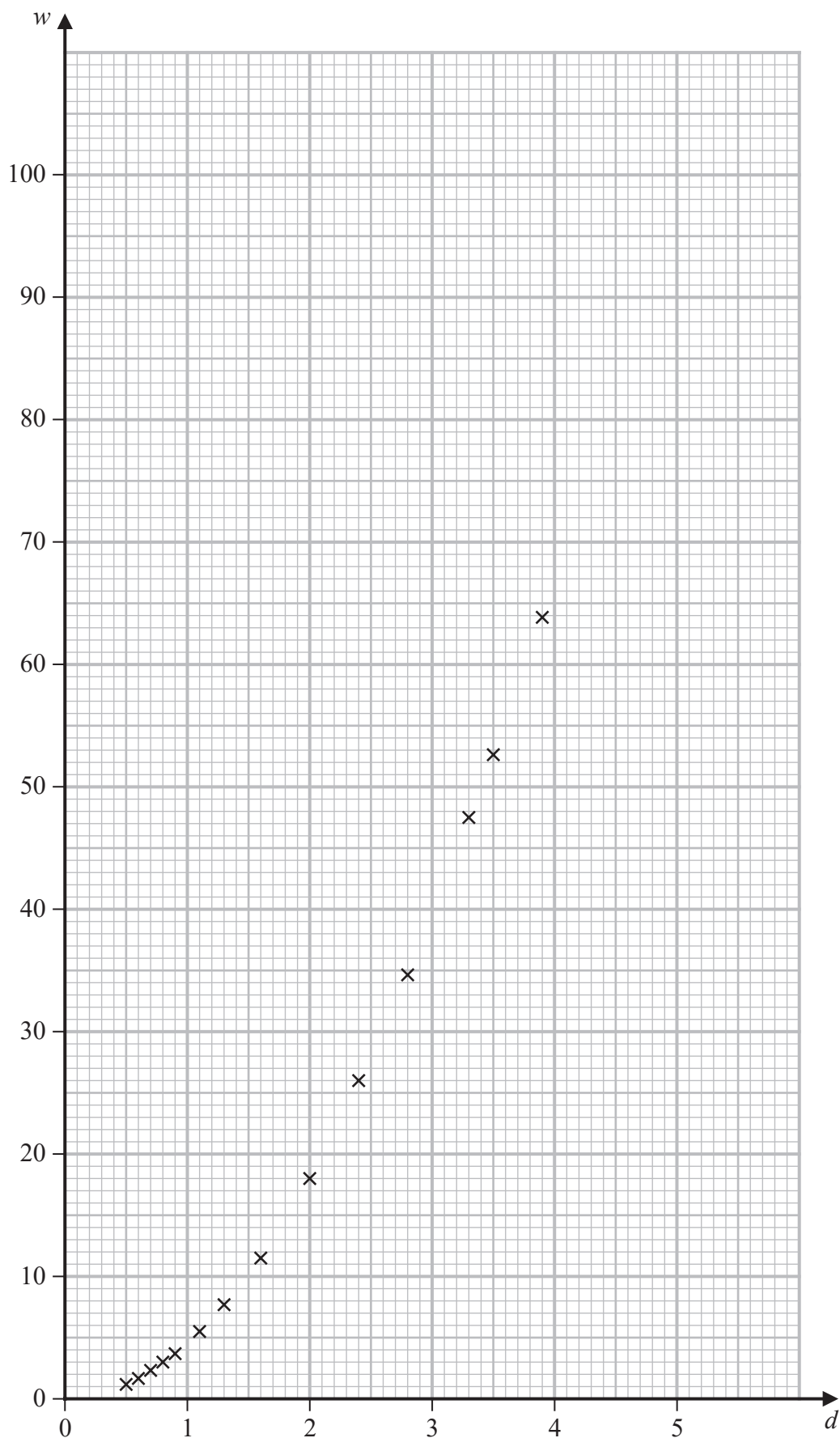
Robert, one of the students, suggests that the model could be improved and intends to find the equation of the line of regression of w on u , where $u = d^2$
He finds the following statistics

$$S_{wu} = 5721.625 \quad S_{uu} = 1482.619 \quad \sum u = 157.57$$

- (e) By considering the physical nature of the problem, give a reason to support Robert's suggestion. (1)
- (f) Find the equation of the regression line of w on u . (3)
- (g) Find the residual sum of squares (RSS) for Robert's model. (2)
- (h) State, giving a reason based on these calculations, which of these models better describes these data. (1)
- (i) Hence estimate the weight at which a piece of wire with diameter 3 mm will fail. (1)



Question 4 continued



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 14 marks)

**TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS
END**



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Thursday 16 May 2019			
Afternoon		Paper Reference 8FM0-24	
Further Mathematics Advanced Subsidiary Further Mathematics options 24: Further Statistics 2 (Part of option G only)			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Bara is investigating whether or not the two judges of a skating competition are in agreement. The two judges gave a score to each of the 8 skaters in the competition as shown in the table below.

	Skater							
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Judge 1	71	70	72	62	63	61	57	53
Judge 2	73	71	67	64	62	56	52	53

Bara decided to calculate Spearman's rank correlation coefficient for these data.

- (a) Calculate Spearman's rank correlation coefficient between the ranks of the two judges. (4)
- (b) Test, at the 1% level of significance, whether or not the two judges are in agreement. (4)

Judge 1 accidentally swapped the scores for skaters *D* and *E*. The score for skater *D* should be 63 and the score for skater *E* should be 62

- (c) Without carrying out any further calculations, explain how Spearman's rank correlation coefficient will change. Give a reason for your answer. (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 10 marks)



2. Lloyd regularly takes a break from work to go to the local cafe. The amount of time Lloyd waits to be served, in minutes, is modelled by the continuous random variable T , having probability density function

$$f(t) = \begin{cases} \frac{t}{120} & 4 \leq t \leq 16 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the cumulative distribution function is given by

$$F(t) = \begin{cases} 0 & t < 4 \\ \frac{t^2}{240} - c & 4 \leq t \leq 16 \\ 1 & t > 16 \end{cases}$$

where the value of c is to be found.

(2)

- (b) Find the exact probability that the amount of time Lloyd waits to be served is between 5 and 10 minutes.

(2)

- (c) Find the median of T .

(2)

- (d) Find the value of k such that

$$\mathbb{P}(T < k) = \frac{2}{3} \mathbb{P}(T > k)$$

giving your answer to 3 significant figures.

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 9 marks)



3. Two students, Jim and Dora, collected data on the mean annual rainfall, w cm, and the annual yield of leeks, l tonnes per hectare, for 10 years.

Jim summarised the data as follows

$$S_{wl} = 42.786 \quad S_{ww} = 9936.9 \quad \sum l^2 = 26.2326 \quad \sum l = 16.06$$

- (a) Find the product moment correlation coefficient between l and w (2)

Dora decided to code the data first using $s = w - 6$ and $t = l - 20$

- (b) Write down the value of the product moment correlation coefficient between s and t .
Give a justification for your answer. (1)

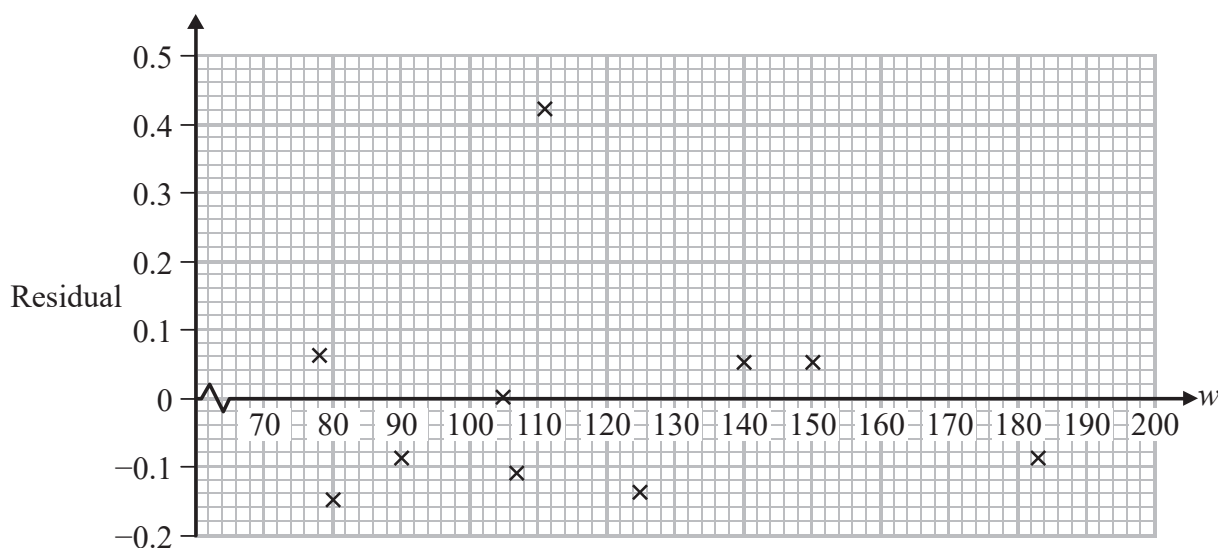
Dora calculates the equation of the regression line of t on s to be $t = 0.00431s - 18.87$

- (c) Find the equation of the regression line of l on w in the form $l = a + bw$, giving the values of a and b to 3 significant figures. (3)

- (d) Use your equation to estimate the yield of leeks when w is 100 cm. (1)

- (e) Calculate the residual sum of squares. (2)

The graph shows the residual for each value of l



- (f) (i) State whether this graph suggests that the use of a linear regression model is suitable for these data. Give a reason for your answer.
(ii) Other than collecting more data, suggest how to improve the fit of the model in part (c) to the data. (2)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 11 marks)



4. The random variable X has a continuous uniform distribution over the interval $[5, a]$, where a is a constant.

Given that $\text{Var}(X) = \frac{27}{4}$

(a) show that $a = 14$

(3)

The continuous random variable Y has probability density function

$$f(y) = \begin{cases} \frac{1}{20}(2y - 3) & 2 \leq y \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

The random variable $T = 3(X^2 + X) + 2Y$

(b) Show that $E(T) = \frac{9857}{30}$

(7)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 10 marks)

**TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS
END**



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Pearson Edexcel
Level 3 GCE

Centre Number

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Further Mathematics

Advanced Subsidiary
Further Mathematics options
24: Further Statistics 2
(Part of option G only)

Thursday 17 May 2018 – Afternoon

Paper Reference

8FM0-24

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. The scores achieved on a maths test, m , and the scores achieved on a physics test, p , by 16 students are summarised below.

$$\sum m = 392 \quad \sum p = 254 \quad \sum p^2 = 4748 \quad S_{mm} = 1846 \quad S_{mp} = 1115$$

- (a) Find the product moment correlation coefficient between m and p (2)
- (b) Find the equation of the linear regression line of p on m (3)

Figure 1 shows a plot of the residuals.

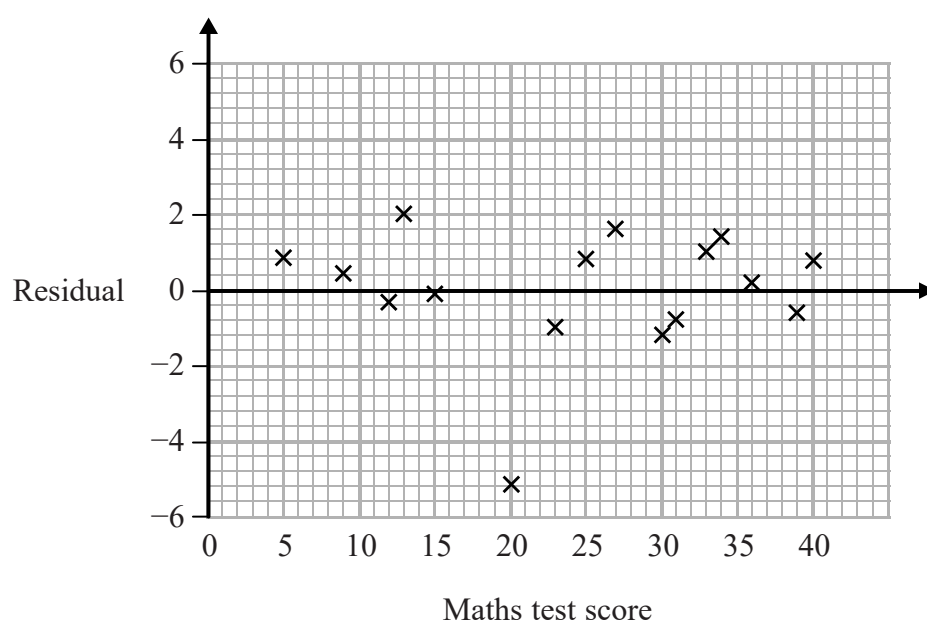


Figure 1

- (c) Calculate the residual sum of squares (RSS). (2)

For the person who scored 30 marks on the maths test,

- (d) find the score on the physics test. (2)

The data for the person who scored 20 on the maths test is removed from the data set.

- (e) Suggest a reason why. (1)

The product moment correlation coefficient between m and p is now recalculated for the remaining 15 students.

- (f) Without carrying out any further calculations, suggest how you would expect this recalculated value to compare with your answer to part (a). Give a reason for your answer. (1)



Question 1 continued

Lined area for writing the answer to Question 1.

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Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 11 marks)



2. The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{8} & 1 \leq x \leq 9 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Write down the name given to this distribution.

(1)

The continuous random variable $Y = 5 - 2X$

- (b) Find $P(Y > 0)$

(2)

- (c) Find $E(Y)$

(2)

- (d) Find $P(Y < 0 \mid X < 7.5)$

(3)

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. The table below shows the heights cleared, in metres, for each of 6 competitors in a high jump competition.

Competitor	A	B	C	D	E	F
Height (m)	2.05	1.93	2.02	1.96	1.81	2.02

These 6 competitors also took part in a long jump competition and finished in the following order, with C jumping the furthest.

C A F D B E

- (a) Calculate Spearman's rank correlation coefficient for these data. (4)
- (b) Stating your hypotheses clearly, test at the 5% level of significance whether or not there is a positive correlation between results in the high jump and results in the long jump. (4)

The product moment correlation coefficient between the height of the high jump and the length of the long jump for each competitor is found to be 0.678

- (c) Use this value to test, at the 5% level of significance, for evidence of positive correlation between results in the high jump and results in the long jump. (2)
- (d) State the condition required for the test in part (c) to be valid. (1)
- (e) Explain what your conclusions in part (b) and part (c) suggest about the relationship between results in the high jump and results in the long jump. (1)



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 12 marks)



4. The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 3 \\ c - 4.5x^n & 3 \leq x \leq 9 \\ 1 & x > 9 \end{cases}$$

where c is a positive constant and n is an integer.

(a) Showing all stages of your working, find the value of c and the value of n

(7)

(b) Find the lower quartile of X

(2)

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 9 marks)

TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS



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Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced Subsidiary

Further Mathematics options

Paper 2G: Further Statistics 1 and Further Statistics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 40 minutes

Paper Reference

8FM0/2G

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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- 6.** The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{18}(11 - 2x) & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find $P(X < 3)$ (2)
- (b) State, giving a reason, whether the upper quartile of X is greater than 3, less than 3 or equal to 3 (1)

Given that $E(X) = \frac{9}{4}$

- (c) use algebraic integration to find $\text{Var}(X)$ (3)

The cumulative distribution function of X is given by

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{18}(11x - x^2 + c) & 1 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (d) Show that $c = -10$ (2)
- (e) Find the median of X , giving your answer to 3 significant figures. (3)

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Question 6 continued

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7. A scientist wants to develop a model to describe the relationship between the average daily temperature, $x^{\circ}\text{C}$, and a household's daily energy consumption, $y\text{kWh}$, in winter.

A random sample of the average temperature and energy consumption are taken from 10 winter days and are summarised below.

$$\sum x = 12 \quad \sum x^2 = 24.76 \quad \sum y = 251 \quad \sum y^2 = 6341 \quad \sum xy = 284.8$$

$$S_{xx} = 10.36 \quad S_{yy} = 40.9$$

- (a) Find the product moment correlation coefficient between y and x . (2)
- (b) Find the equation of the regression line of y on x in the form $y = a + bx$ (3)
- (c) Use your equation to estimate the daily energy consumption when the average daily temperature is 2°C (1)
- (d) Calculate the residual sum of squares (RSS). (2)

The table shows the residual for each value of x .

x	-0.4	-0.2	0.3	0.8	1.1	1.4	1.8	2.1	2.5	2.6
Residual	-0.63	-0.32	-0.52	-0.73	0.74	2.22	1.84	0.32	f	-1.88

- (e) Find the value of f . (2)
- (f) By considering the signs of the residuals, explain whether or not the linear regression model is a suitable model for these data. (1)

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Question 7 continued

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[illegible]

8. The continuous random variable X is uniformly distributed over the interval $[-3, 5]$.

(a) Sketch the probability density function $f(x)$ of X .

(2)

(b) Find the value of k such that $P(X < 2[k - X]) = 0.25$

(3)

(c) Use algebraic integration to show that $E(X^3) = 17$

(3)

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Question 8 continued

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(Total for Question 8 is 8 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Specimen Paper			
		Paper Reference 8FM0/24	
Further Mathematics Advanced Subsidiary 24: Further Statistics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Question 1 continued

(Total for Question 1 is 5 marks)



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2. Charlie and Diane are trying to model the continuous random variable X . Charlie suggests modelling the cumulative distribution function of X with $F_C(x)$ where

$$F_c(x) = \begin{cases} 0 & x < 2 \\ \frac{1}{6}(9x - x^2 - 14) & 2 \leq x \leq 5 \\ 1 & x > 5 \end{cases}$$

- (a) Explain what is wrong with Charlie's model. (2)

Diane suggests modelling the probability density function of X with $f_D(x)$ where

$$f_D(x) = \begin{cases} k \left(x^2 - 2x + \frac{2}{3} \right) & 2 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

- (b) Find the value of k for Diane's model. (3)

- (c) Sketch $f_D(x)$ for Diane's model. (2)

- (d) Use algebraic integration on Diane's model to find $E(X)$. (3)

Diane is going to calculate the median of X .

- (e) With reference to the shape of your sketch in part (c), state, giving a reason, whether or not she should expect the median of X to be smaller, equal to or greater than $E(X)$. (2)



Question 2 continued

Handwriting practice area with 30 horizontal lines.



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 12 marks)

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4. Naasir is studying the hearing function of workers in his factory. He takes a random sample of 10 workers from his factory. He records their hearing function as a percentage, h , where 100% indicates perfect hearing. He also records the number of years they have worked in the factory, w .

The results are summarised below.

$$\bar{h} = 86.5 \quad \bar{w} = 10.5 \quad S_{hh} = 198.5 \quad S_{ww} = 470.5 \quad r = -0.620$$

- (a) Stating your hypotheses clearly and using a 5% level of significance, test whether or not there is evidence of a negative correlation between hearing function and the number of years worked in Naasir's factory. (3)

- (b) State why your test in part (a) is consistent with the use of a linear regression model to describe the relationship between hearing function and the number of years worked in Naasir's factory. (1)

Naasir wishes to estimate the effect that the number of years spent working in his factory has on the hearing function of his workers. He decides to create a linear regression model.

- (c) Using the above data find Naasir's linear regression model. (6)

- (d) Use this model to estimate the effect on hearing function of working in Naasir's factory for 5 years. (1)

- (e) Calculate the residual sum of squares for this model. (2)

Erica was one of the workers included in the sample. She is 58 years old, has worked in the factory for 10 years and her hearing function is 80%

- (f) Calculate the residual for Erica's data. (2)

A friend visited Naasir's factory and suggested that all new workers should have an assessment of their hearing function when they start working in his factory.

- (g) (i) State, giving your reasons, a limitation of Naasir's current model.
(ii) Explain how he could use this extra information to design an improved model. (2)



Question 4 continued

Lined area for writing the answer to Question 4 continued.



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Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 17 marks)

TOTAL FOR FURTHER STATISTICS 2 IS 40 MARKS

